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Data Analytics

Research Assignment 1

Answers

14 Sep 2024

Section A : Data Fundamentals

1) A database is an organized, structured way to store data electronically. It is designed in a way that data/information can be accessed, managed and retrieved easily.

Databases are essential in a modern world because they enable organizations to handle high volumes of information efficiently and in a secure manner.

Real world examples

→ In healthcare, An electronic Health Record System is used by hospitals to using relational databases like Epic or Cerner to store patient demographics, histories, medication lists and lab results.

→ TIER Net (Three Interlinked Electronic Register) is a national database used primarily in healthcare clinics in South Africa. It serves as a tracking database for HIV, Tuberculosis and Antiretroviral Therapy patient records.

2) A datawarehouse is a centralized, large-scale database that is designed for querying, reporting and for data analysis. It organizes historical data pulled from many sources.

A regular database is ~~for~~ ^{optimized} for fast, real-time daily operations.

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A company would use a data warehouse as it is optimized for analyzing large volumes of historical data across an entire organization.

3) Data Lake → It is a centralized storage

repository designed to hold massive volumes of raw data in

its native, unorganized format

until it is needed. A data lake ingests everything as it is.

A business would use Data Lakes over a data warehouse

when its operations require flexibility with

unstructured data, advanced capability and low

state cost scalability.

4) Data Mart → A specialized, focused subset of

data warehouse designed to

serve the specific reporting

and analytical needs of a

single business unit or department

While a data warehouse pulls data across the

entire business into one massive

repository, a data mart isolates a portion

of that enterprise data to make access

faster and simpler for a single department

or target group.

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5) • **Structured data** → Rigid, pre defined model (table with rows and columns) eg SQL; Spreadsheets (Excel)

• **Semi structured** → Flexible tags, without rigid schema eg No SQL databases.

• **Unstructured data** → Completely unorganized, no pre defined data model or schema eg Data Lake; File servers.

6) **Data Modeling** is a process of creating a visual diagram or blueprint that defines how data is structured, stored, organized and related within an information system.

Data modeling is essential when designing a system as it ensures data quality and integrity, it also improves query performance and scalability.

Examples

→ **Conceptual Data Model**: Designed for business stakeholders and system architects.

→ **Physical Data Model**: This model is for database administrators and developers. Details how the model will actually be built in a specific database system.

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7) Data Mining → It is the process of analyzing large datasets to uncover hidden patterns, trends, relationships and anomalies that are not

obvious through simple observation.

Companies use it to predict customer / consumer behaviour and to optimize operations and pricing.

8) A Database Management System (DBMS)

→ A software that serves as an interface between database and its end users or applications.

It provides administrative tools to store, manage, retrieve, update and secure data, whilst maintaining integrity and performance.

Four examples

1) MySQL → Open-source relational Database MS that uses structured query language (SQL). It organizes data into rigid tables of rows and columns.

It is used for E-commerce sites,

2) PostgreSQL → An advanced - open source object relational database system known for its strict adherence to SQL standards.

It is used for Geospatial services and

⑤ and financial systems.

3) Oracle database

→ A commercial, high performance relational DBMS built for large-scale enterprise computing.

It is built for critical operations.

Such as banking transactions.

4) MongoDB

→ A leading, non-relational NoSQL document

database. Instead of rows and tables, it

stores data as flexible, schema-less

JSON-like documents.

It's suitable for mobile apps and

real-time content catalogs.

9) The Three Steps of ETL:

E - Extract: Data is extracted from various sources.

Such as operational SQL databases,

CRM platforms (Salesforce) and payment

gateways.

T - Transform: The raw, unorganized data undergoes

a series of operations to align it

with business logic and analytical

needs. Removes duplicate records,

fixes structural errors and handling

missing values, and standardizing

formats.

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L-Load: The newly transformed, high-quality data is written into ~~data~~ ^{data} ~~target~~ ^{target} ~~repository~~ ^{repository}. A

Importance of ETL in data analytics,

✓ Eliminate data silos - Companies use a dozen of apps. ETL pulls data out of ~~isolated~~ ^{isolated} systems and merges it into a single Unified ~~data~~ ^{analytical} hub.

✓ Ensures Data Quality and Accuracy

Raw data is filled with errors, null values and duplicates. The transform step ensures the data is clean.

✓ Optimize Query performance

Querying production database directly for reports can slow down active applications. ETL offloads analysis to structured warehouse, designed for rapid querying.

10) CSV: Comma-Separated Values

→ A plain-text, row-based format that stores

table data. Each line represents a record

and each value is separated by commas or

delimiters.

It is ideal to use for simple data exchange.

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JSON (Java Script Object Notation)

→ A text-based, semi-structure format that stores data in hierarchical key-value pairs and arrays.

Ideal to use for web API responses, mobile app data logs.

Parquet

→ An open source, columnar, binary storage format designed for efficient analytical processing within big data ecosystems.

It is ideal for a large-scale data warehouse, data lakes and heavy analytical / OLAP queries where systems regularly aggregate specific columns.

Avro

→ A row-based binary data format developed under the Apache Hadoop project. It stores a light weight JSON schema directly alongside compact binary data.

It is ideal for high streaming data pipelines, event logging and real-time data ingestion.

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Section 8

Artificial Intelligence and Machine Learning

11) AI is a field of computer science dedicated to building software systems and machines that can perform tasks that typically require human intelligence. These tasks include learning from experience, reasoning, problem-solving, recognizing speech and patterns, and understanding natural language.

Narrow AI (Weak AI)

↳ Designed and trained for to perform a specific ^{single} task or limited set of related tasks. It operates within a defined framework and can't apply its intelligence to tasks outside its domain, even if it performs its specific tasks faster or better than human.
eg → Email spam filters, Netflix recommendations algorithm.

General AI (Strong AI)

↳ A theoretical form of AI that possesses human-level cognitive capabilities across any domain. It would be able to learn, reason, adapt, plan and solve problems across environments, including unfamiliar tasks, without needing specialized training.

eg → Science fiction — Systems that can think, reason, feel and make any decision.

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12) Machine Learning

↳ A computer science, data science and artificial

intelligence subset that enables systems to

learn and improve from data without

additional programming intervention.

Machine Learning vs Traditional Programming



System takes both input

data and known output.

(or patterns) to generate

its own algorithms

and rules. The model

adjusts its internal logic

Automatically

automatically as it processes scenario, changes the

new data and code must be updated

manually.

Three main types of machine learning:

• Supervised learning: Models train on labeled data

to where input & correction to

known target variable.

common uses: linear regression.

• Unsupervised learning: Algorithms analyze unlabeled

data to discover hidden

patterns, groups or

structural relationships

without human intervention

or prior targets.

common uses: Anomaly

detection.

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Reinforcement learning is an autonomous agent, learns optimal behaviour through trial and error in a dynamic environment, eg. Robotics control and autonomous navigation.

13) Deep learning is a specialized subset of machine learning that relies on multi-layered artificial neural networks to automatically extract features and learn complex patterns from vast amounts of data without explicit human engineering.

The relationship between AI, Machine Learning and DL is that AI is the broadest umbrella then ML is a specific branch of AI focused on building algorithms that learn from data to make predictions or decisions. Lastly, DL is a specialized subset of ML. Traditional ML models can plateau in accuracy as data grows, DL models continue to improve with more data and can compute power automatically by discovering the key patterns on their own.

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14) Natural Language Processing (NLP)

→ A branch of artificial intelligence that enables computers to understand, interpret, manipulate and generate human language. It bridges the gap between human communication (which is complex and unstructured) with structured computer code (which is strict and structured) by using techniques from computational linguistics, machine learning and deep learning.

Everyday realworld applications

- Voice Assistants (Siri, Google Assistant, Alexa)
- Smart compose and Autocorrect (Gmail, Smartphone keyboards)
- Language Translation Services (Google Translate)

15) Large Language Model (LLM)

→ A type of AI deep learning model trained on massive amounts of text data.

Difference between a LLM and standard machine learning model is in scope and flexibility.

Standard machine learning is usually built for narrow specific tasks like predicting house prices, classifying spam. An LLM is a general purpose model capable of performing dozens of various language tasks like writing code, translating languages, summarizing essays.

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Two well-known LLMs and their use cases

→ GPT-4 (OpenAI) is a multimodal, large language model designed to power conversational assistants like ChatGPT, process complex reasoning tasks, draft written content and analyze uploaded files.

→ Claude 3.5 Sonnet (Anthropic) is a frontier LLM widely used for long-context analysis, complex document processing, coding assistance and technical writing and research.

(b) Generative AI (GenAI) is a subfield of AI focused on creating new content such as text, images, computer code, audio or video.

How it works: GenAI typically works through three main stages:

- 1) Training on Massive Dataset
- 2) Learning patterns and Relationships
- 3) Predictive Generation

Tools that use GenAI

- ChatGPT
- Midjourney
- GitHub Copilot.

17) Training Dataset is the initial set of data used to teach AI or Machine Learning model how to recognize patterns, make predictions or perform tasks.

It is important to ensure high quality training data is used in order to get optimal results.

To ensure accuracy and generalization, high quality (well-labeled) data allows the model to learn genuine underlying rules rather than random noise.

A sophisticated neural network architecture cannot fix bad input. If the training data contains errors, missing information or outdated facts, AI will internalize these flaws and execute incorrect predictions.

If the training data is biased or inaccurate, this can result in algorithm bias/discrimination. Tools such as facial recognition can have a flawed preference.

In healthcare, clinical AI assistants can provide dangerous misinformation that can produce misdiagnosis and false ~~prognosis~~ ^{produce} confidence, produced instead of falsehoods.

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18) Computer vision is a field of AI that enables computer and ^{software} systems to receive, process and derive meaningful information from digital images, videos and other visual inputs.

How Machines see and interpret images

→ To a computer, a digital image is not a picture of a face, car or landscape but rather a grid of numbers representing individual fixed pixels.

Real world cases

-) Autonomous vehicles
-) Medical Imaging Diagnostics in healthcare

Section C: API, Tools, & Developer concepts.

19) API (Application Programming Interface) is a set of defined rules, protocols and mechanisms that allows two different software applications to communicate and exchange data with each other. It acts as a bridge or messenger between systems without requiring them to know how the underlying code on either side actually works.

Example in real world, books and books API

Modern health systems use a standardized API framework called Fast Healthcare Interoperability Resources. When a person logs into Apple

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Health App on their iPhone, the app sends a ^{Secure} request via the hospital's FHIR App then the hospital's server verifies your identity and grants permission. The API then pulls your lab results, allergies and prescriptions straight out the hospital database into your phone.

Why APIs are important in Modern Software and Data systems

- Prevents Reinventing the wheel : Developers do not need to build every service from scratch.
- Enables Interoperability & System Integration
- Enhances Security and Controlled Access
- Power Automation & AI Infrastructure

20) An API Key is a unique alphanumeric string that an application or user passes to an API, to identify and authenticate the caller.

The purpose of an API key is for verification, authorization and Access Control, Usage Tracking and Rate limiting.

In real world, API monitors call volumes to enforce quota limits, charge usage fees or prevent service abuse.

API Key should be stored and protected by:

- Use Environment Variables & keep keys out of your source code.

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→ Use secret tools: (Store keys in secret tools such as Google secret Manager, AWS secret Manager.

→ Never embed keys in front-end code: Do not put API keys in client-side code

→ Restrict key super

→ Rotate keys periodically

If keys are exposed publicly, it can result in financial loss, data breach and service disruption & ad exhalitation.

21) A REST API is an application programming interface that follows a specific set of architectural rules for designing web services.

REST → Representational State Transfer

Main HTTP Methods used in REST APIs

1) GET (Read): Retrieves data from the server.

It is a read-only operation.

Excluding GET request

should never modify, create,

or delete "data" on the

["data" on server]: "data"

} "data"

2) POST (Create): Submits new data to the server to create a new resource.

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3) PUT (Update/Replace): Replaces an existing resource entirely or creates it if it doesn't exist yet,

4) DELETE: Removes a specific resource from the server.

22) JSON stands for JavaScript Object Notation.

It is a lightweight, text-based data interchange

Format used to store, exchange structured information between client and a server.

JSON is commonly used for API and Data Exchange.

Exchange:

- Human and Machine Readable
- Lightweight and Fast.
- Native Web ^{Compatibility} ~~compat~~
- Flexible and Hierarchical

JSON Object:

"user_id": 1658,

"first_name": "Lindiwe",

"is_active": true,

"email": "lindiwe.radebe@discoveryhealth.com",

"roles": ["data analyst"]

"address": {

"city": "Johannesburg"

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20) Prompt engineering is the practice of designing, structuring and refining text inputs to guide large model such as ChatGPT and Claude towards generating accurate, relevant, high quality and properly formatted outputs.

It is an important skill as it save time and token usage, well-structured prompts ~~eliminate~~ basic and form follow-ups. It also unlocks high level reasoning.

For example of a prompt:

→ Write an email seeking a refund for a flight

[Example prompt]

This prompt does not provide the airline details or details of the flight like date, flight number.

Improved version:

→ Write an email seeking a refund for a flight from FlySafair, the flight was booked for 19 December 2026, flight booking reference number is #02301ZXTV17.

24) MAP Function is a higher-order function in programming and data processing that takes an existing collection (like a list or array) and applies a specific transform function to every single element in that collection, returning a brand new collection containing transformed items.

What it does in Data Processing

- Data Transformation: Convert raw data into a new format
 - Extracting Fields: Pulls out a specific attribute from a list of complex objects
- Parallel Processing: Each element is processed independently of others

Example to double every number in a list

JavaScript

```
const numbers = [1, 2, 3, 4];
```

// Using array .prototype .map() with an arrow function

```
const doubled = numbers.map(x => x * 2);
console.log(doubled); // Output [2, 4, 6, 8]
```

25) Cloud computing is the on the demand delivery of computing services including servers, storage, databases, networking, software and analytics, over the internet or a private network.

Cloud Platform

1) IaaS (Infrastructure as a Service)

Provides a hardware and software framework

~~fundamental virtualized computing~~

infrastructure over the internet

On which developers can build, run, and manage custom applications. The cloud provider handles all backend infrastructure, operating system updates, server maintenance and scaling allowing developers to focus entirely on writing application code.

Eg AWS Elastic Beanstalk.

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2) IaaS (Infrastructure as a service)

→ Provides fundamental infrastructure over the internet.

You ~~rent~~ ^{rent} raw, hardware components such as

Virtual servers, network connections and

Storage spaces.

Eg Amazon EC2 or Google Compute

Engine.

3) SaaS (Software as a service)

→ Provides fully functional, end-user software

application over the web. The provider

manages everything - from underlying physical

servers and software code to security

patches and application updates.

Eg Google Workspace or Salesforce.

2b) Version Control System is a software tool that tracks, records and manages changes made to source

code, scripts, dataset and project files over time. It

keep a complete record of who made the

changes, when, ^{and} why allowing you to

recall or revert to specific previous

versions whenever needed.

Git is the world's most popular open-source

distributed version control system. It is

widely used by developers and data analysts for

- Safety and error recovery

- Seamless Collaboration

- Experimentation without risk

- Auditability and Traceability

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Commit → A saved snapshot of your project's file at a specific point in time. When you make changes to files and create a commit, Git saves a record of those specific line-by-line changes along with a unique ID and log message describing what was altered.

Branch → An independent timeline of development that splits off from the main project thread. It allows you to develop new features, fix bugs or experiment with alternate analytical models in an isolated workspace. Once your work is tested and verified, you can merge it back to primary project line.

27) A Jupyter Notebook is an open source web based interactive computing environment that allows users to create and share documents containing live code, execution outputs, formatted text, equations and visualization in a single document.

Primary language is not commonly used with Python though it supports R and Julia.

It is popular because of interactive code execution, integrated documentation and data exploration and storytelling.

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28) Python is a high level, interpreted, general purpose programming language known for its clean, readable syntax and versatility.

Python is popular because it is readable and simple. Its syntax closely resembles plain English, significantly lowering the learning curve and allowing ~~and~~ analysts to focus on solving data problems rather than complex code mechanics.

It also offers a massive ecosystem, unparalleled collection of pre-built open source libraries specifically designed for data manipulation, statistics and machine learning.

Furthermore it provides strong community and integration; it easily integrates with databases, web applications, cloud infrastructure and big data systems, making it industry standard across tech companies.

Four common python libraries

- **Panda** : Primary library for data manipulation and analysis. Provides flexible data structure.
- **Numpy** : Numerical Python - provides support for large multi-dimensional arrays and matrices along a collection of high performance mathematical functions.

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3) Matplotlib - Fundamental plotting and data visualization library. Allows data analysts to generate static, animated interactive visualization

Scikit Learn - The standard machine learning library in python. Provides

Simple efficient tools for predictive data analysis, including algorithms for classification, regression, clustering and model evaluation

Section D

Data Ethics, Governance & Careers

29) Data privacy refers to the branch of data protection concerned with the proper handling, processing, storage and use of personal information. It focuses on individuals' autonomy - ensuring that individuals maintain control over how their personal identifiable information such as names, addresses, financial details, ^{stored} and ~~stored~~ ^{collected} and managed by third parties.

Data privacy is ^{important} ~~important~~ as it protects individual autonomy and freedom, prevents financial loss and security threats, prevents discrimination and unfair profiling and establishes Business trust and accountability.

GDPR is General Data Protection Regulation, it is a comprehensive data privacy law imposed by the European Union. It became effective 25 May 2018. It establishes strict standards for how organizations collect, store, process and transfer personal data belonging to individuals.

GDPR is applicable to:

- EU member states (27 countries) ^{e.g. Germany, Poland, France, Spain, Ireland}
- European ^{Economic} Area states, includes non EU members
- United Kingdom ^{UK}
- Global Extraterritorial Reach: e.g. South Africa, USA, Japan or Australia

30) Data Ethics is a branch of ethics that evaluates the moral implications of collecting, processing, storing, sharing, and using data. It's like, as the AI model is built upon it. It goes beyond simple legal compliance, focusing on fairness, transparency, accountability, privacy and avoiding social harm.

Three Examples of Unethical data usage.

-) An automated recruitment platform uses AI to score candidate resumes. The training data comes from 10 years of company hiring records which heavily favoured male applicants. As a result the model specializes resumes containing men's colleges.

This is problematic as it automates and scales historical systemic bias under the guise of objective software. Qualified candidates are discriminated against based on gender.

Unsanctioned Scraping of Artists' work for Generative AI

A generative AI company scraped billions of copyrighted digital paintings, photography and original artwork from independent artists off public websites without obtaining consent or paying royalties.

This is problematic as it exploits intellectual property and livelihoods.

→ Hyper-targeted Behavioral Manipulation

A data analytics company harvested personal profile data from tens of millions of social media users without explicit consent. They built detailed psycholo^{with customized}gical profiles to target voters ~~on~~ polarizing political misinformation aimed at manipulating election outcomes.

It is problematic as it violates user consent and personal autonomy.

3.) Data Governance is a system of rights, processes, roles, standards and metrics designed to manage an organization's data assets throughout their entire data ^{life} cycle.

Key components of good data governance framework

1) Role and Accountability structure: Define clear roles and responsibilities for managing data assets across business.

2) Data quality management: Establish metrics and routines to maintain high data integrity.

3) Data ^{Architecture} standards and data management.

4) Data privacy, security and compliance: Implement policies to safeguard sensitive information and ensure regulatory compliance.

5) Change management and organizational culture

Ensure data governance is accepted as an ongoing business rather than a one-time IT project. Fostering an environment that promotes data literacy and trains employees to see data as a respected and key strategic asset.

32) AI bias occurs when machine learning model produces systematic, prejudiced outputs due to erroneous assumptions, flawed algorithms or skewed data used during its development.

When an AI model is biased, it does not evaluate information neutrally. Instead, it systematically favours certain outcomes, groups or individuals over others.

Real world Example : Brazil's Predictive Policing algorithm.

- Use of Predictive Policing software used in the United States. The algorithm was trained on historical arrest data. Because lower-income and minority neighbourhoods had historically been disproportionately over-policed, the underlying training data reflected higher arrest rates in these specific areas.

Potential consequence is erosion of public trust?

When citizens realise automated software perpetuates racial or socioeconomic profiling, trust in both judicial and technology erodes.

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	Data Analyst	Data Engineer	Scientific Data Engineer	AI Engineer
Role	Inspect, clean and analyze historical data to help organizations understand performance, trends to answer specific business questions.	Focuses on infrastructure, quality and governance. They oversee data architecture, pipelines, security policies and regulatory compliance to ensure data safety and integrity.	Focuses on future. They build predictive algorithms, machine learning models and statistical experiments.	Focuses on product strategy and business integration.
Skills	- SQL querying - Data cleaning - Data visualization - Business domain knowledge	- ETL - Data security - Data lineage - Vendor management	- Machine learning - Advanced statistics - Software engineering	- Product management - Data ethics
Tools	- SQL - Power BI - Excel - Python - Tableau	- SQL Server - Snowflake - Azure Data Tools	- Python - SQL - Jupyter Notebook - Databricks	- Jira/Confluence - Open AI APIs - Power BI - Git/Github

34) Business Intelligence comprises the strategies, technologies and software practices used by organizations to collect, integrate, analyze and present raw data into actionable insights.

Two popular BI tools:

1) Power BI is a cloud based business analytics service that connects to hundred of data sources to create interactive dashboards and reports. It helps business data decision making by integrating with Microsoft environment and leverage automated data refreshes. It allows decision makers to track key performance indicators in real time.

2) Tableau by sales force

It is a high performative visual analytics platform renowned for its advanced data visualization capabilities and drag and drop interface.

Tableau excels at handling massive data sets to visually surface hidden patterns, correlations and outliers.

Executive teams use it to build drag and drop exploratory dashboards enabling rapid scenario analysis.